

◆ 2.1 Identity & Access Management (IAM)

✓ What it does

- Login, registration, authentication
- Role-Based Access Control (RBAC)
- Audit tracking

⚙️ Backend Working

- User logs in → JWT token generated
- Roles determine access (Citizen, Judge, etc.)
- Every action logged in AuditLog

⚙️ Tech Stack

- Spring Boot
- Spring Security (JWT)
- Hibernate + JPA
- MySQL

💡 Example

- Judge can access only hearings, not admin configs

◆ 2.2 Citizen Registration & Profile Management

✓ What it does

- Register users
- Upload ID proofs
- Profile management

⚙️ Backend Working

- Store citizen data
- Upload documents (store file path in DB)
- Verification workflow

⚙️ Tech Stack

- Spring Boot REST APIs
- JPA (Entity: Citizen, CitizenDocument)
- File Storage (Local / Cloud)

💡 Example

- Aadhaar/PAN upload and verification

◆ 2.3 Case Filing & Document Management

✓ What it does

- File legal cases
- Upload evidence
- Track case status

⚙️ Backend Working

- Citizen/Lawyer submits case
- Clerk verifies
- Status flows: Filed → Under Review → Approved

⚙️ Tech Stack

- Spring Boot (Microservice: Case Service)
- Spring Data JPA
- Hibernate Relationships
 - Case → Many Documents

💡 Example

- FIR-like system for courts

◆ 2.4 Hearing Scheduling & Court Proceedings

✓ What it does

- Schedule hearings
- Track proceedings
- Assign judges

⚙️ Backend Working

- Clerk schedules hearing
- Judge updates proceeding notes
- Status updated per hearing

⚙️ Tech Stack

- Spring Boot
- JPA Relationships
 - Case → Hearing → Proceeding
- Scheduler (Optional: Spring Scheduler)

💡 Example

◆ 2.5 Judgment & Order Management

✓ What it does

- Judges give verdict
- Issue legal orders
- Track compliance

⚙️ Backend Working

- Judge submits judgment
- Stored with case reference
- Linked to compliance module

⚙️ Tech Stack

- Spring Boot
- JPA
- Transactional Management

💡 Example

- Final court decision + order
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◆ 2.6 Compliance & Audit Management

✓ What it does

- Track if orders are followed
- Perform audits
- Store findings

⚙️ Backend Working

- Compliance records created
- Audit reports generated
- Linked to case/hearing/judgment

⚙️ Tech Stack

- Spring Boot
- JPA
- Batch Processing (Spring Batch optional)

💡 Example

- Check if court order was implemented
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◆ 2.7 Reporting & Analytics

✓ What it does

- Dashboards
- KPIs (case resolution time, etc.)

⚙️ Backend Working

- Aggregate DB queries
- Generate reports
- Expose APIs for frontend charts

⚙️ Tech Stack

- Spring Boot
- Custom SQL / JPA Queries

💡 Example

- Monthly case resolution report
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◆ 2.8 Notifications & Alerts

✓ What it does

- Send alerts (SMS, Email, In-app)
- Notify about hearings, judgments

⚙️ Backend Working

- Event triggers → Notification service
- Queue-based (optional for scale)

⚙️ Tech Stack

- Spring Boot
- Spring Events
- Email/SMS APIs

💡 Example

- “Your hearing is scheduled tomorrow”